

Scenario

A network operator is using ONOS and wants to integrate a new type of legacy network device that uses a proprietary protocol.

Question: Which ONOS interface (Northbound, Southbound, Eastbound, Westbound) would be most relevant for adding support for this new device's proprietary protocol? Explain why.

Answer:

Relevant Interface: Southbound Interface

Explanation:

- **Role of the Southbound Interface:** The Southbound interface in SDN architectures (including ONOS) handles communication between the SDN controller and underlying physical or virtual network devices (data plane switches, routers, and legacy appliances).
- **Protocol Adaptation via Drivers and Providers:** ONOS uses modular *Providers* and *Drivers* within its Southbound abstraction layer to translate high-level controller commands into specific control protocols. To support a legacy device using a proprietary protocol, the operator must implement a custom Southbound driver and provider plugin.
- **Abstracting Data Plane Heterogeneity:** The Southbound abstraction converts generic core ONOS primitives (such as flow rules, port configurations, or device discovery) into the exact frame/packet formats required by the proprietary protocol, allowing ONOS to manage legacy hardware alongside open standards like OpenFlow, NETCONF, or gNMI.